

Multi-Layer SOC Intelligence Pipeline

Log Normalization (log_parser.py)

Convert messy logs → structured security data

Convert logs into SIEM-compatible structure.

->ECS Mapping (**Elastic Common Schema style**)

Logs are transformed into an ECS-like (Elastic Common Schema) structure to ensure compatibility with SIEM systems such as Elasticsearch

HYBRID ENGINE (Rule Engine & LLM + RAG)

Your system works in **two layers of intelligence**:

1. RULE ENGINE (First Layer — Fast Detection)

Purpose : Detect obvious known attacks instantly **without AI**.

2. LLM + RAG (Second Layer — Intelligence Layer)

This is used when the rule engine is **not confident or no match is found**.

LLM (Ollama / Llama3)

Purpose : Understand **complex or ambiguous logs**

Why this hybrid design is important

Advantages:

1. Speed

- Rules = instant detection

2. Accuracy

- LLM = handles complex cases

3. Reduced hallucination

- RAG = grounded knowledge

4. SOC realism

This is how real SOC systems work:

- Splunk rules
- SIEM correlation
- ML/AI enrichment layer

RAG (ChromaDB + MITRE ATT&CK) — “Give the LLM brain knowledge”

RAG = **context injection** so the LLM doesn't guess.

Multi-Log Attack Campaigns + Correlation

This is where your project becomes **real SOC-level**

Role

Detect attacks that span **multiple logs**, not just one.

Because:

One log = weak signal

Many related logs = real attack

Correlation means: don't analyze logs individually — analyze them together to see the full attack

Threat Intelligence (TI Enrichment)

Role

Check if IP / domain is **known malicious**

Why TI matters

- external validation
- reduces false positives
- increases confidence

SOAR (Response Automation)

Role

Automatically suggest or trigger actions.

SOAR should NOT blindly execute

It should **recommend or simulate actions**

Based on:

- severity
- confidence
- correlation
- TI

Hallucination Guard (Reliability Layer)

The hallucination guard module ensures output reliability by validating LLM-generated alerts against raw log data. It enforces entity grounding (IP, URL, user), logical consistency, and evidence verification. A rule-based scoring mechanism assigns a hallucination risk score based on detected inconsistencies.

Evaluation Layer (Alert Quality Scoring System)

The evaluation layer computes a weighted quality score for each alert based on 12 SOC-relevant dimensions including detection accuracy, MITRE mapping, severity calibration, TP/FP correctness, and hallucination absence. The system produces a normalized score out of 10 to assess alert reliability and operational usefulness.

Feedback Loop (Continuous Learning Layer)

The feedback loop module introduces human-in-the-loop learning where SOC analysts label alerts as true or false positives. These labels are stored and used to dynamically adjust scoring biases per attack category, improving future alert classification accuracy.